Jacob Bacus

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EDUCATION

Dartmouth College, Hanover, NH **June 2027**

Bachelor of Arts, Major in Computer Science and Middle Eastern Studies **GPA 3.87/4.0**

Relevant Coursework: Object-Oriented Programming, Foundations of Applied CS, Software Design & Implementation, Principles of Programming Languages, Machine Learning & Statistical Analysis, Discrete Math, Algorithms, Web Development

Activities: Contextual Dynamics Laboratory, Class Council, Zeta Psi Fraternity President, Morocco Language Study Abroad, Glee Club

RELEVANT EXPERIENCE

Klaviyo, Boston, MA **June 2026-August 2026**

Software Engineering Intern

- Scoped and developed a new system for auditing account activity. Used by customer support and developers 1000+ times monthly.
- Contributed agentic log filtering to an ongoing project to provide AI support to customers analyzing account activity.
- Created multiple custom AI tools, including skills and custom extensions for workflows, to help accelerate developers at Klaviyo.

DALI Lab, Hanover, NH **February 2025-Present**

AR/VR & Game Design Domain Lead *(April 2026-Present)*

- Manage developers in one of the Lab's core pillars across 10 active project teams.
- Direct mentorship, developer experience, and project delivery initiatives while contributing to development for company partners.

Fullstack Developer *(January 2026-Present)*

- Worked with design teams through agile development to ship fullstack web applications for clients from approved designs.
- Created an AI-first education platform allowing educators to make courses with LLM integrations such as a course design agent.
- Developed an AI literacy platform for the Dartmouth CS Department including full LLM-chatbot functionality for explanations.

AR/VR Developer *(February 2025-April 2026)*

- Coordinated with design and hardware teams on simulated reality projects in an agile environment.
- Developed an admin interface and hardware code for an eye-tracking exhibit at the Salvador Dali Museum in St. Petersburg, FL.
- Designed wearable prototypes using sensors, LEDs, and textiles to create garments that react to the environment.

P1 Construction, Lenexa, KS **March 2025-June 2026**

Software Engineer (Remote) *(June 2025-June 2026)*

- Helped update company web services with strong API contracts and custom Revit data integrations.
- Maintained and updated codebase of custom Revit Tools including refactoring to a shared plugin structure.
- Developed new software functionality for designers based on usage data and feedback from previous implementations.

Software Engineer Intern *(March 2025-June 2025)*

- Built a suite of custom Revit tools to assist Virtual Design and Construction workers through automated part placement and alteration.
- Worked directly with designers to develop and iterate upon features that reduced friction through intuitive UI.
- Tested and documented tools to ensure longevity and better enable future development.

PROJECTS

Twigby (Sample Course) **January 2026-June 2026**

- Launched a platform to help educators integrate AI into their everyday coursework.
- Created a block-based system for developing custom courses.
- Created a LangGraph-based agent to design and edit full courses autonomously for users.

GroupTrip.ai (Demo Video) **February 2026-March 2026**

- Developed a web application with React/TypeScript frontend and MongoDB backend where users can collaboratively plan trips.
- Integrated AI agent functionality allowing an agent to autonomously find and suggest real-world flights, activities, and hotels.
- Iterated on product to create collaborative features such as trip chat, group polls, finances, interactive map, and export functionality.

Creative Navigator (Email for Further Demo) **March-June 2025**

- Collaborated individually with a University of Kansas professor to develop an AI assistant for classroom use.
- Led user interviews to define core features and establish an MVP surrounding audio journaling.
- Used a React frontend, Supabase Database, and serverless functions to store conversations, integrate with LLMs, and implement RAG.

DALI Museum Exhibit (Demo Video) **September-November 2025**

- Utilized Tobii Nano Pro to track eye movement during user viewing of exhibit art.
- Established an admin dashboard to allow museum administrators to maintain the exhibit.
- Developed research workflows to help curators understand how comments affect the viewing of paintings.

SKILLS AND INTERESTS

Technical: Object-Oriented Programming, Algorithm Design, Numerical Techniques, Predictive Modeling, Applied AI, API Integration, Programming Language Design, Dynamic Programming, Graph Theory, Software Design Principles, Agile Methodology, Supabase, RAG, Prompt Engineering, Skill Creation, Agent Harnesses, Unity3D, Revit

C#, C, Java, Python, JavaScript, TypeScript, HTML, CSS, React, Tailwind CSS, SQL, ANTLR, Git, Bash

Professional: Team Leadership, Project Coordination, Project Scoping, Cross-Functional Collaboration, Mentorship

Languages: English, Arabic (conversational)

Interests: Software Engineering, Machine Learning, Data Engineering, Middle Eastern Studies, Snowboarding, Hiking, Travel